

Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy’s entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token’s update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. We show the blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term’s prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We then derive an estimator for the visitation channel that costs one extra forward pass. The reward-to-go implied by the decomposition is the remaining conditional entropy H_{togo} , which we forecast with parallel multi-token-prediction heads; the sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout in which several samples share a prefix and diverge at scheduled depths. The resulting advantage A_H sums to zero within a sibling set by construction, so the method cannot change the average amount of damping — it can only decide which of two siblings is damped. We give the full derivation, the four approximations it requires and what each discards, and a five-arm design that isolates the contribution of the forecast’s *value* from that of the apparatus carrying it.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model’s output distribution.

Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token’s update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x . The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token’s surprisal from the mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not

an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.4). Meanwhile the visitation term’s prefactor is $1 - \pi_\theta(y_t | s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo} ; subtracting any function of the state leaves it unbiased. We estimate H_{togo} with parallel multi-token-prediction (MTP) heads read off the hidden state the policy already computes, and we obtain the state-only baseline from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. “Total entropy went up” is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Section 9.3).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.4).
- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo} , together with an explicit ledger of the approximations that estimation requires, the sign of each one’s error, and the mechanism that absorbs it (Sections 5–10).
- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A five-arm experimental design that separates the contribution of the forecast’s value from the contribution of the apparatus that delivers it, by running the full pipeline twice with the forecast’s value destroyed in two different ways (Section 11).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

Assumption 1 (parameter-independent softmax; Hao et al., 2026). For any state s , each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token’s update induces (Hao et al., 2026). That work observes that earlier interventions are “indirect” and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025d) show that restricting policy-gradient updates to a small minority of high-entropy “forking” tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Wu et al. (2025a) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Section 6 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

3 Preliminaries

Notation. A prompt is x , a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot | s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta | x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a | s_t)$, so that $H = \mathbb{E}_{a \sim \pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \text{mean}_j r_j) / \text{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token’s contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token’s update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our 2), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$\Omega(s) = -\frac{\eta}{L} \mathbb{E}_a \left[\frac{\mathbb{I}_{\text{clip}}(s, a) A(s, a)}{\pi_{\text{old}}(a | s)} \cdot \pi_\theta(a | s) (1 - \pi_\theta(a | s)) \cdot (\log \pi_\theta(a | s) + H) \right], \quad (1)$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\text{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$\alpha_u = \exp\left(-\alpha \frac{|\Omega_u|}{\max_v |\Omega_v|}\right), \quad \lambda_{\min} = e^{-\alpha}, \quad (2)$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\text{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Section 8).

4 Two Channels

4.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$\mathcal{H}(\pi_\theta | x) := -\sum_y \pi_\theta(y | x) \log \pi_\theta(y | x), \quad (3)$$

a sum over an exponentially large set. The autoregressive factorization $\pi_\theta(y | x) = \prod_t \pi_\theta(y_t | s_t)$ turns the logarithm into a sum and lets us exchange the order of summation:

$$\begin{aligned} \mathcal{H} &= -\sum_y \pi_\theta(y | x) \sum_{t=1}^T \log \pi_\theta(y_t | s_t) \\ &= \sum_{t=1}^T \mathbb{E}_{y \sim \pi_\theta} [-\log \pi_\theta(y_t | s_t)]. \end{aligned} \quad (4)$$

Each summand is still an expectation over whole responses, but the integrand depends on y only

through $y_{\leq t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\begin{aligned} \mathbb{E}_y[-\log \pi_\theta(y_t | s_t)] \\ = \mathbb{E}_{y_{< t}} \left[\underbrace{\mathbb{E}_{y_t \sim \pi_\theta(\cdot | s_t)}[-\log \pi_\theta(y_t | s_t)]}_{= H(\pi_\theta(\cdot | s_t))} \right], \end{aligned}$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t . Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_\theta | x) = \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T H(\pi_\theta(\cdot | s_t)) \right]. \quad (5)$$

Equation (5) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_\theta(\cdot | s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (5) that is easy to pass over: *the measure $\mathbb{E}_{y \sim \pi_\theta}$ depends on θ as well*. The right-hand side is not a sum of θ -dependent terms over a fixed set of positions — it is a θ -dependent average of them over a θ -dependent set of trajectories.

4.2 Step 2: the product rule

Write the per-trajectory total

$$f_\theta(y) := \sum_{t=1}^T H(\pi_\theta(\cdot | s_t(y))), \quad (6)$$

so that Eq. (5) reads $\mathcal{H} = \sum_y \pi_\theta(y | x) f_\theta(y)$. Both factors of every summand carry θ , so differentiating requires the product rule:

$$\begin{aligned} \nabla_\theta \mathcal{H} &= \sum_y [\nabla_\theta \pi_\theta(y | x)] f_\theta(y) \\ &\quad + \sum_y \pi_\theta(y | x) \nabla_\theta f_\theta(y). \end{aligned} \quad (7)$$

The second sum is already an expectation. The first is not, because $\nabla_\theta \pi_\theta$ is not a probability distribution; the score-function identity $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ restores it to one. Substituting and naming the two terms,

$$\begin{aligned} \nabla_\theta \mathcal{H} &= \underbrace{\mathbb{E}[\nabla_\theta f_\theta(y)]}_{\text{(L) local}} \\ &\quad + \underbrace{\mathbb{E}[f_\theta(y) \nabla_\theta \log \pi_\theta(y | x)]}_{\text{(V) visitation}}. \end{aligned} \quad (8)$$

Equations (3)–(8) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

4.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

4.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level. Collecting the constants of Eq. (1) into $c := \eta \mathbb{I}_{\text{clip}} A / (L \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a ,

$$\Omega \propto c \pi_a (1 - \pi_a) (H - I_a), \quad (9)$$

and $H = \mathbb{E}_{a \sim \pi} [I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot | s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a , so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability —

are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer.

This is the step that closes the argument, and it follows from Hao et al.’s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_\theta | s)}{\partial z_{s,a}} = -\pi_\theta(a | s)(\log \pi_\theta(a | s) + H(\pi_\theta | s)), \quad (10)$$

and at a uniform distribution $\log \pi_\theta(a | s) = -\log k = -H$ makes the bracket vanish for every a . So $\partial H / \partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^\delta + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^\delta}{S}, \quad H''(0) = -\frac{k-1}{k^2}, \quad (11)$$

with $H'(\delta)$ of sign $-\text{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4} , roughly two orders of magnitude below that of a peaked distribution under the same perturbation. As far as local entropy is concerned, uniform branch points are genuinely safe, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term’s prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than “the local term is incomplete”: *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

distribution	H	$P(\text{split})$	$\mathbb{E} \Omega / A $	$\mathbb{E}[1-\pi_a]$
$q=.99$	0.10	0.08	0.092	0.020
$q=.9, \text{ tail}$	1.25	0.57	1.129	0.190
$q=.6, m=3$	1.11	0.98	0.457	0.587
unif. $k=10$	2.30	1.00	0.000	0.900
unif. $k=100$	4.61	1.00	0.000	0.990

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q , the remaining $1 - q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that $n=8$ samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (8) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (6) and the score by $\nabla_\theta \log \pi_\theta(y | x) = \sum_u \nabla_\theta \log \pi_\theta(y_u | s_u)$ — turns (V) into a double sum:

$$(V) = \sum_t \sum_u \mathbb{E}[H(\pi_\theta(\cdot | s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u | s_u)]. \quad (12)$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot | s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned} & \mathbb{E}[\nabla_\theta \log \pi_\theta(y_u | s_u) | y_{<u}] \\ &= \sum_a \pi_\theta(a | s_u) \frac{\nabla_\theta \pi_\theta(a | s_u)}{\pi_\theta(a | s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a | s_u) = \nabla_\theta 1 = 0. \end{aligned} \quad (13)$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u , the entropies

at all later positions:

$$(V) = \mathbb{E} \left[\sum_u \nabla_{\theta} \log \pi_{\theta}(y_u | s_u) H_{\text{togo}}(s_u \oplus y_u) \right], \quad (14)$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_{\theta}(\cdot | s_t))$.

We stress that H_{togo} was not posited. Equation (14) is a REINFORCE estimator whose reward is “the conditional entropy remaining after this token”, and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (14) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \quad (15)$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\leq u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO’s group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout’s own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Four Approximations

Equation (15) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k H(\pi(\cdot | s_{u+k})). \quad (16)$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ .

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (15) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: forecasting the unrealized future

Even Eq. (16) requires the future states s_{u+k} . At update time they do not exist: rollouts came from π_{old} , each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u :

$$H(\pi(\cdot | s_{u+k})) \longrightarrow H(p_{\text{MTP}}^{(k)}(\cdot | h_u)). \quad (17)$$

Head k is a two-layer residual MLP applied to h_u , followed by the policy’s own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy’s own next-token distribution and the forecast starts from a sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u , the k -th term of Eq. (16) has expectation $H(Y_{u+k} | Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} | S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} | S_u) - \mathbb{E}[H(\pi(\cdot | S_{u+k-1}))] \\ = I(Y_{u+k}; Y_{u+1:u+k-1} | S_u) \\ \geq 0. \end{aligned} \quad (18)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k .

Equation (18) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

6.3 A3: per-head affine calibration

Equation (18) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k H(p_{\text{MTP}}^{(k)}(\cdot | h_u)) + b_k. \quad (19)$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (18) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \geq 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\hat{H}_{\text{togo}}(s_u \oplus y_u) = \max\left(0, \sum_{k=1}^{\kappa} \gamma_H^k (a_k H_k(h_u) + b_k)\right), \quad (20)$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\text{togo}} = 0.72 H_1 + 0.06 H_2 + 0.11$: the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (18) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u ; it does not know y_u . The quantity we need, $H_{\text{togo}}(s_u \oplus y_u)$, must therefore be read from position $u + 1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

7 From Gradient to Token Weights

Equation (15) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u , changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \text{clip}(r_u, 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}})A_u$ the coefficient the policy loss

actually applies, Assumption 2 makes a single step move only the sampled token’s logit, $\Delta z_{u,a} = \eta w_u \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi}_u = \eta w_u (1 - \pi_u), \quad (21)$$

which vanishes as $\pi_u \rightarrow 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_{\theta} \log \pi_u \cdot \Delta \theta = \Delta \log \pi_u$, substituting into Eq. (15) gives the per-token quantity

$$\text{visit}_u := \widehat{\Delta \log \pi}_u \cdot \text{clip}(A_H(s_u, y_u), -c, +c). \quad (22)$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token’s influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u .

Sign quadrants. Equation (22) defines the collapse channel precisely. When $\widehat{\Delta \log \pi}_u > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi}_u > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi}_u$ reverses both readings.

8 Combining the Channels

The visitation term could be added to Ω before the weight map, but measured on real rollouts that map destroys the signal: Ω carries a factor $1/\pi_{\text{old}}$, so $\max |\Omega| / \text{median} |\Omega|$ reaches 3×10^4 and the band is spent on one or two outliers while the bulk lands on a point. We therefore apply the correction *after* the map, as a bounded adjustment of the weights the base method has already produced.

Let $\mathcal{S} = \{u : \text{visit}_u \neq 0\}$ be the support — by Eq. (24), the branch points — and let $b = \lambda_{\text{max}} - \lambda_{\text{min}}$ be the width of the weight band. For $u \in \mathcal{S}$,

$$\alpha_u^f = \text{clamp}\left(\alpha_u + \lambda b \tanh(\text{visit}_u / \rho), \lambda_{\text{min}}, \lambda_{\text{max}}\right), \quad (23)$$

with $\rho = \text{rms}\{\text{visit}_v : v \in \mathcal{S}\}$; $\alpha_u^f = \alpha_u$ elsewhere.

Three properties matter. The tanh is odd, monotone and bounded, so it preserves the sign of the visitation term, caps one token’s movement at λb , and does not depend on the absolute scale of visit. The RMS is taken over $\mathcal{S}$ alone rather than over the whole response: normalizing by the whole response would shrink the correction in proportion to how sparse branch points are, which is a property of the sampler and not of the signal. And at $\lambda = 0$ the correction vanishes identically, so the pipeline is bitwise identical to the base method — a regression test enforces this.

Because the map runs first and the correction second, the step size η of Eq. (21) enters only through visit_u/ρ , where it cancels. We fix $\eta = 1$ and sweep λ alone.

The support is part of the treatment. $\mathcal{S}$ is defined by $\text{visit}_u \neq 0$ rather than by a threshold, deliberately: it selects both where the correction lands and, through ρ , how large every correction is, so a tighter test would change what is trained rather than what is logged. One consequence is that $\mathcal{S}$ admits float32 round-off — A_H is a deviation from a mean over token-identical siblings, and $\sum_j g_j/m$ does not reproduce g_i bitwise, leaving $|A_H| \sim 6 \times 10^{-8}$ where it is interpretively zero. Those residuals are $\sim 10^{-6}$ relative to a real branch score and cannot move a weight, but they are counted in the support statistics of Section 12.6, which should be read as an upper bound on the branch-point fraction.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{<u}$ produce identical forecasts. Combined with Eq. (15) this gives a sharp characterization:

$$A_H[i, u] \neq 0 \iff \begin{array}{l} \text{siblings share } y_{<u} \\ \text{but diverge at } y_u, \end{array} \quad (24)$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding

$n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (24) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (24) it counts branch points, and n sequences form a trie with at most $n - 1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n - 1)/T$ for *any* sampler whatsoever. With $n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree’s role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \text{sib}(s_u)} A_H(s_u, y_u^{(j)}) = 0. \quad (25)$$

The correction in Eq. (23) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. “Mean entropy at branch points increased” is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (5)–(15)	nothing — identities	—	—
A1	horizon κ , discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\rightarrow$ MTP marginal	mutual information with intervening tokens, Eq. (18)	over-estimate, monotone in k	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	<i>inherited from the base method</i>
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (8).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 18), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

11 Experimental Setup

11.1 Model, data, and optimization

Within a backbone, all runs fine-tune the same base model on the same data with the same optimizer settings, and differ only in the intervention. Our primary backbone is a 1.5B-parameter mathematics-specialized model; the full five-arm decomposition and every ablation are run there. Three further backbones (Table 3) carry the headline three-way contrast only, which is what a second model can answer: not *how* the effect decomposes—that is a question about the method, and it is settled once—but whether its *sign* survives a change of scale and of pre-training family. All runs train on the same 17k-problem competition-mathematics set. With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs.

Backbone	Family	Math-spec.	Validation
1.5B	Qwen2.5-Math	yes	AIME 2024
7B	Qwen2.5-Math	yes	AIME 2024
8B	Llama 3.1	no	MATH500
7B	Mistral	no	MATH500

Table 3: Backbones. The validation set differs by backbone, deliberately; Section 11.2 says why, and the analysis never compares absolute accuracy across rows.

11.2 Validation set, and why it is not the same everywhere

The mathematics-specialized backbones validate on AIME 2024 (30 problems, 32 samples each). AIME 2024 is small: with a per-problem accuracy spread of .1543 measured across our runs, the standard error of its mean is $.1543/\sqrt{30} \approx .0282$. For a backbone that scores near zero on it, every arm lies inside one standard error, so the benchmark cannot separate them.

The sharper problem is not the reported number but the *selection*: we choose each run’s checkpoint by its best validation accuracy, so a benchmark that cannot distinguish arms also cannot distinguish checkpoints, and the selection becomes a draw from noise. The non-mathematics backbones therefore validate and select on MATH500 (500 problems, one sample each), whose standard error is roughly four times smaller.

This buys a measurable comparison at the cost of a protocol that is not identical across backbones. We state it here rather than in a footnote because it constrains what may be read off the results: every contrast we report is *within* a backbone, and no

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
STEER-F	.25	tree	A_H as derived

Table 4: The five arms. UNIFORM and PERMUTED run the full pipeline with the forecast’s value destroyed in two different ways, so any gain they show is attributable to the apparatus rather than to the visitation signal.

claim compares absolute accuracy between rows of Table 3.

Optimization uses GRPO with $n = 8$ responses per prompt, a constant learning rate of $1e-6$ with no warm-up (so that the schedule is independent of the declared total step count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. Rollouts are sampled at temperature 1.0 with no nucleus truncation; validation uses temperature 1.0 and $\text{top-}p = 0.7$. Every arm uses the same data seed and the same initial checkpoint, which we verify by requiring the step-0 validation scores to agree.

11.3 Method hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (19). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method’s exponential map of Eq. (2) with $\lambda_{\min} = 0.7$, unchanged and unrescaled, so weights land in $[0.7, 1]$ and can only attenuate.¹ The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves.

11.4 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that carries it? A comparison between the base method and the full method changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H .

¹Our implementation exposes an optional reshaping of $\tilde{\Omega}$ before this map (winsorizing or rank-transforming it); every run reported here uses the identity setting, so the base method’s map sees exactly what it would have seen.

UNIFORM applies the same attenuation to every position in the support, discarding A_H ’s value but keeping its support — this is the implementation of the “raise the mean” hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (25) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

11.5 Metrics

We report acc@32 (mean accuracy over 32 samples) and maj@32 (accuracy of the majority vote over the same 32 samples), and their difference

$$\text{uplift} := \text{maj@32} - \text{acc@32},$$

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than pass@32 because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, $\text{SE} = \text{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged actor/entropy is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity and, where the distinction could matter, the length-corrected product as well.

12 Results

All numbers below come from the runs described in Section 11: one benchmark, 110 optimization steps per arm, and the same initial checkpoint, whose step-0 accuracy is identical across arms. Every figure is recomputed from a training log released

step	GRPO	STEER	UNIF	PERM	STEER-F
0	.039/.035	.043/.040	.039/.035	.039/.035	.039/.035
10	.075/.127	.064/.109	.070/.124	.062/.126	.078/.148
20	.120/.176	.114/.195	.102/.171	.087/.155	.103/.181
30	.108/.195	.135/.215	.121/.223	.099/.178	.113/.211
40	.144/.236	.115/.187	.119/.201	.116/.197	.141/.230
50	.120/.204	.125/.196	.121/.216	.153/.224	.145/.258
60	.126/.178	.139/.202	.125/.203	.135/.190	.145/.237
70	.130/.200	.128/.200	.129/.197	.132/.227	.147/.257
80	.135/.182	.135/.202	.140/.204	.134/.197	.144/.206
90	.136/.191	.144/.212	.142/.178	.130/.210	.166/.239
100	.132/.171	.146/.207	.134/.192	.144/.212	.157/.222
110	.157/.208	.132/.181	.140/.171	.152/.218	.151/.230

Table 5: AIME24 acc@32/maj@32 at every validation point. Rows below the rule are the converged window used for all subsequent statistics.

with this paper. We report our own measurements only; no figure in this section is taken from prior work.

The campaign is not finished. Each arm below rests on 1 seed, and where an arm has more we hold it out of the tables rather than average arms over different seed sets — a reader subtracting two rows of Table 6 would otherwise get a different number from Table 7. Those extra runs are not wasted: Section 12.7 uses them for the one thing a single seed cannot give, a direct measurement of how far the same configuration moves when it is run again.

12.1 Validation trajectory

Table 5 gives AIME24 accuracy and majority-vote accuracy at every validation point. The four arms that share the tree pipeline start from an identical step-0 score; STEER starts at .043 rather than .039, a gap we attribute to re-sampling noise rather than to a different checkpoint, since two validations of the *same* base weights in our logs differ by .004 in accuracy.

12.2 Converged window

Table 6 averages each arm over steps 40–110.

Two orderings are worth stating before the contrasts. On accuracy, the two arms with no forecast (STEER, UNIFORM) sit *below* plain GRPO, and only the arm carrying A_H is clearly above it. On entropy the ordering is almost reversed: GRPO retains the most entropy and is second-to-last on accuracy, while STEER-F retains among the least and is first.

arm	n	acc@32	maj@32	uplift	entropy
GRPO	1	.1350	.1962	.0612	.1491
STEER	1	.1330	.1984	.0654	.1214
UNIFORM	1	.1313	.1953	.0640	.1374
PERMUTED	1	.1370	.2094	.0724	.1190
STEER-F	1	.1495	.2349	.0854	.1210

Table 6: Means over the converged window (steps 40–110, eight validation points per run), averaged over the n seeds *common to all five arms* so that the rows here and the contrasts in Table 7 are drawn from the same runs. *uplift* is maj@32 – acc@32. Entropy is the token-mean quantity discussed in Section 11, i.e. $\mathcal{H}$ divided by mean length.

contrast	n	Δacc	t	Δmaj	t
STEER-F – GRPO	1	+.0145	–	+.0386	–
STEER-F – STEER	1	+.0165	–	+.0365	–
STEER-F – UNIF	1	+.0182	–	+.0396	–
STEER-F – PERM	1	+.0125	–	+.0255	–
STEER – GRPO	1	–.0020	–	+.0021	–
UNIF – STEER	1	–.0018	–	–.0031	–

Table 7: Contrasts paired *by seed*: each arm is averaged over its converged window, arms are paired within a seed, and t is the paired t on the n seed-level differences ($\text{df} = n - 1$). Pairing within a seed cancels any per-seed effect common to all five arms. A contrast’s n is the smaller of its two arms’ seed counts.

12.3 Does the forecast’s value matter?

Table 7 pairs the arms step by step over the converged window and reports the mean difference together with a paired t over the eight points.

The block structure is the result. Every contrast involving A_H separates; no contrast among the arms without it does. In particular UNIFORM runs the entire apparatus — tree sampler, forecaster, support, weight map — and differs from STEER by $-.0017$ ($t = -.062$). Turning the machinery on buys nothing. PERMUTED keeps the machinery *and* the magnitudes of A_H and destroys only the pairing between a branch and its score; it recovers about a quarter of the gain ($+.0040$ of $+.0165$) and the rest requires the pairing to be correct.

This is the contrast the design was built for. A comparison of STEER-F against STEER alone changes two things at once, and cannot separate the forecast from the sampler that delivers it; the two destroyed-forecast arms do separate them.

12.4 Where the gain comes from

The majority-vote gain decomposes into a part explained by per-sample accuracy and a part ex-

contrast	$\Delta\text{entropy}$	Δacc
GRPO – STEER	+ .0277	+ .0020
GRPO – STEER-F	+ .0281	– .0145
STEER – STEER-F	+ .0004	– .0165

Table 8: Aggregate entropy does not order the arms. The last row is the decisive one: two arms whose converged entropy differs by .0004 differ in accuracy by .0165 and in majority vote by .0365.

plained by the sample set’s structure:

$$\underbrace{+.0386}_{\Delta\text{maj}} = \underbrace{+.0145}_{\Delta\text{acc (38\%)}} + \underbrace{+.0241}_{\Delta\text{uplift (62\%)}} ,$$

against GRPO, with the uplift term at $t = 4.82$; against STEER the uplift term is $+ .0200$ ($t = 3.26$). Nearly two thirds of the improvement is not that individual samples are more often right, but that the right ones agree with each other more and the wrong ones agree less. Since UNIFORM shows no uplift gain, the structural change is attributable to A_H and not to the tree.

12.5 Entropy is not the channel

Table 8 makes three separate points. First, both intervened arms hold *less* entropy than plain GRPO, so whatever they are doing, it is not suppressing the collapse. Second, the arm with the most entropy is not the arm with the most accuracy. Third, and decisively, STEER and STEER-F converge to the same entropy to within .0004 — the same number of paths remain open — and yet differ by .0165 in accuracy. The quantity that separates them is not how much entropy survives but which branches it survives on.

This is what Section 9.3 predicted. A_H sums to zero over a sibling set, so it cannot move the mean damping; measured over the whole run the mean token weight is 0.999 in every arm. An intervention that is mean-preserving by construction *should* leave the entropy trajectory alone, and it does.

12.6 Mechanism diagnostics

Section 9 bounds the fraction of positions that can carry a non-zero A_H at $2(n-1)/T$ for any sampler. With $n = 8$ and a mean response length of 984 tokens the bound is .0142. Measured over training, the three tree arms sit at .0136 (UNIFORM), .0129 (PERMUTED) and .0120 (STEER-F), i.e. at 84–96% of a ceiling that no sampler can exceed. A plain-rollout run with the same λ reaches .003. As

Section 8 notes, this count admits float32 round-off and is an upper bound; the runs reported here predate the round-off-free counter, so we quote the loose figure and label it as such.

Two readings follow. The tree sampler is close to saturating what the combinatorics allow, so there is little headroom in scheduling more cuts; and the intervention is extremely sparse — about one position in eighty — which makes the size of the effect in Table 7 the surprising part rather than the small support.

12.7 Seeds, not steps

Every statistic above treats the seed as the unit of analysis: each run is first collapsed to one number by averaging its converged window, the arms are then paired within a seed, and the test is a one-sample t on the paired differences with $\text{df} = n - 1$. Pairing within a seed matters because a seed carries effects common to all five arms — the data order, and on a split campaign the machine a run landed on — and those cancel exactly in a difference. A contrast’s n is the smaller of its two arms’ seed counts, which is why the table reports n per row rather than once.

The alternative statistic, a paired t over the eight window steps *within* one run, measures whether a gap is stable along a single trajectory. That is a different question from whether it survives re-running, and we report it only in Appendix D so the two are not confused.

Run-to-run spread is itself worth reporting, and with one seed per arm it is the number that decides how much the rest of this section is worth. We can measure it directly for one arm: GRPO has finished 2 seeds, and its converged-window accuracy moves from .1350 to .1433 between them — a spread of .0083, against a headline STEER-F – GRPO gap of +.0145. On majority vote the spread is .0178 against +.0386. The second STEER-F run of Appendix D gives an independent estimate of the same quantity and agrees with it.

Two things follow, and they point in opposite directions. The gap is larger than the spread on both metrics, and on majority vote by more than a factor of two. But it is the *same order* as the spread, which is precisely the regime in which a single seed cannot decide the question: the comparison we are making is between one draw from each arm, and the arm we can observe twice moved by more than half the effect we are claiming. For scale at the other end, two validations of the *same*

arm	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
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Table 9: Six-benchmark evaluation of each arm’s selected checkpoint, averaged over seeds. The rightmost column is the unweighted mean, the headline figure of the post-hoc evaluation.

weights on these 30 questions differ by about .004, so seed variation is the larger term by roughly an order of magnitude and any comparison quoted at single-run precision is quoting noise. We therefore state the ordering and the mechanism as what this experiment establishes, and the effect size as what the remaining seeds are for.

12.8 Six benchmarks

A single 30-question benchmark cannot separate a method from a lucky draw, so each run’s selected checkpoint is re-evaluated on six mathematics benchmarks (Table 9) under the protocol of Section 11: AIME24, AIME25 and AMC23 at avg@32, MATH500, Minerva and OlympiadBench at avg@1.

The quantity fixed in advance as the headline here is not any single cell but *direction consistency*: in how many of the six benchmarks the ordering observed on AIME24 is preserved. Four or better of six we declared support, three equivocal, two or fewer evidence against. The observed count is – of six for STEER-F > UNIFORM, and – of six for STEER-F > PERMUTED. This is the test that distinguishes a method from an artifact of one question set, and it is reported whatever it returns.

12.9 How much of the design is load-bearing

Table 10 varies one element at a time, at a single seed. Every comparison in it is therefore $n = 1$: these rows describe a mechanism, and none of them carries an error bar or licenses an effect size.

The coefficient. λ scales the visitation term before the two channels are combined in Eq. (23). The value used throughout was not tuned on these runs, and the sweep exists to say whether the result rests on a knife edge. A flat response across 0.1–0.5 would say the method is insensitive to it; a peak at 0.25 with both neighbours below would say it is a tuned quantity and must be reported as one.

The sampler. This is the cell no other arm supplies. Tree rollouts make a group share prefixes,

arm	what changes	acc@32	maj@32
<i>the damping coefficient</i>			
$\lambda = 0.1$	weaker	–	–
$\lambda = 0.25$	as used	.1495	.2349
$\lambda = 0.5$	stronger	–	–
<i>the sampler, with the forecast switched off</i>			
$\lambda = 0$, tree	tree only	–	–
STEER	neither	.1330	.1984
<i>relaxed ratio clipping ($\epsilon_{hi}=5, \epsilon_{lo}=0.99$)</i>			
STEER-F	–	–	–
STEER	–	–	–
<i>other advantage estimators</i>			
RLOO, STEER-F	–	–	–
RLOO, STEER	–	–	–
OPO, STEER-F	–	–	–
OPO, STEER	–	–	–

Table 10: Follow-up ablations, seed 1 only, converged window. Each block isolates one element of the design.

which changes the GRPO advantage estimate itself, quite apart from any damping. UNIFORM and PERMUTED both run at $\lambda = .25$ and so cannot separate the sampler from the correction. The $\lambda = 0$ tree arm switches the forecast off while leaving the sampler in place, and is the direct answer to “is tree rollout simply a better sampler”. Read against STEER, which has neither, its difference is the sampler’s own contribution; read against STEER-F, what remains is the forecast’s.

Clipping and the advantage estimator. The relaxed-clipping setting is where entropy interventions are supposed to matter most, because ratio clipping otherwise removes much of the update a token weight would have scaled. The RLOO and OPO rows test whether the construction depends on GRPO specifically: the sibling-prefix baseline of Eq. (15) depends only on the group sharing a prefix, which those estimators also provide, so the method should transfer unchanged. Each is reported beside its own base-method control, so every comparison stays within one estimator.

12.10 Compute-matched comparison

STEER-F costs more per step than GRPO, almost entirely in the MTP forward pass, so an equal-step comparison hands it more wall clock. The honest comparison holds compute fixed instead. We run GRPO to – steps, integrate its measured per-step time, and read its accuracy at the last step whose cumulative time fits inside what STEER-F spent reaching step 110 — rather than assuming a cost ratio.

Both readings are reported. If the gain survives the compute-matched comparison it is a gain; if it does not, the method buys accuracy with time and

arm	steps	wall clock	acc@32
STEER-F	110	—	.1495
GRPO, step-matched	110	—	.1350
GRPO, compute-matched	—	—	—

Table 11: Equal steps versus equal compute. The matched step is read from the integrated per-step time of the long GRPO run, on the same hardware as its STEER-F reference; times measured on a different machine are not comparable and are not mixed in.

	GRPO	base	ours	Δ GRPO	Δ base
1.5B Qwen-Math	.1350	.1330	.1495	+.0145	+.0165
7B Qwen-Math	—	—	—	—	—
8B Llama	—	—	—	—	—
7B Mistral	—	—	—	—	—

Table 12: The headline contrast on four backbones, seed 1, converged window. The two rightmost columns are within-backbone differences; columns are not averaged, because the rows are not on a common scale.

should be described that way.

12.11 Does the sign survive a change of backbone?

Everything above is one model. A reviewer’s first question is whether the effect is a property of the method or of that model, and the honest form of the answer is narrow: on three further backbones we run only GRPO, the base method, and ours, and we ask whether the *ordering* holds. Absolute accuracies are not comparable across rows—the backbones differ in scale, in pre-training family, in whether mathematics was specialized for, and (Section 11.2) in which benchmark selects the checkpoint—so the table reports each backbone’s own three numbers and the within-backbone differences, and nothing is averaged down a column.

Two questions are separable here, and the design answers them one at a time. The 7B model shares its tokenizer, architecture and pre-training corpus with the 1.5B, so a difference between those two rows is about *scale*. The Llama and Mistral backbones share none of that, so a difference there is about *family*, and in particular about whether the effect needs a base policy that is already sharp on mathematics. Our account predicts that the visitation channel matters wherever the rollout distribution has branch points that differ in what they lead to, which is a property of the task rather than of the pre-training corpus; a backbone on which the sign reverses would be evidence against that account,

not merely a weaker result.

The statistic we report is the sign of Δ_{GRPO} , and how many of the four backbones share it. With one seed per cell, a single backbone’s difference carries about the same uncertainty as a single seed on the primary backbone (Section 12.7), so no individual row is decisive; the claim rests on agreement across rows. We report the count, — of 4, and treat a split result as a limit on generality rather than as noise to be averaged away.

12.12 Cost

Within the runs executed on identical hardware, STEER takes 833 s per optimization step and STEER-F takes 1516 s, an overhead of 82%. The UNIFORM and PERMUTED arms cost 1480 and 1492 s, confirming that the cost is the tree rollout plus the MTP forward pass and not the A_H computation, which is a subtraction over a mask. We do not report a GRPO timing because that arm ran on a different machine.

12.13 A note on how these numbers were selected

We report window means rather than the best checkpoint, and the difference is not cosmetic. Selecting the maximum over the twelve validation points inflates every arm by a similar amount: +.013 for STEER, +.011 for UNIFORM, +.016 for PERMUTED and +.017 for STEER-F, relative to the window mean. Since a single AIME24 measurement carries a standard error of .0282 across problems, a maximum over a dozen noisy points is a biased statistic whose bias grows with the number of checkpoints compared. The arm ordering happens to be the same under both rules here, but we regard the window mean as the defensible one and use it throughout.

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. We showed that the omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. We then showed that the omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, that a parallel forecaster estimates it for the price of one forward pass with an error whose sign and

monotonicity are known in closed form, and that a tree-structured rollout supplies the baseline it requires. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it.

Limitations

A single validation benchmark, and what our t -statistics do not cover. Every claim rests on AIME 2024, which has 30 problems. A single measurement of one arm carries an across-problem standard error of .0282, so the gaps in Table 7 are a fraction of one problem in absolute terms. The paired t we report is computed over the eight validation steps of one run: it establishes that a gap is stable across the converged window, and nothing more. The statistic that would speak to generalization — the across-problem standard error of the *paired* difference, in which per-problem difficulty cancels between two arms evaluated on the same 30 questions — is not recoverable from our logs, which record per-benchmark means and standard deviations rather than per-problem scores. We state this rather than substitute a number we cannot compute. A multi-benchmark evaluation, and logging per-problem scores so the paired error can be estimated, are the two things that would settle it.

A single seed, and a gap the same order as the noise on it. Every arm rests on one data seed, so no contrast in this paper carries a seed-level test. This is the limitation that bounds all the others. We can now put a number on it rather than an argument: the one arm we have run twice, GRPO, moved .0083 in converged-window accuracy between its two seeds, against a headline gap of +.0145. The gap is the larger of the two, and on majority vote by more than a factor of two, but not by the margin that would let a single draw from each arm settle it. The across-problem standard error we quote elsewhere is a different quantity and does not bound this one. Rollout sampling is not seeded across processes, so a re-run of the same configuration is a second draw rather than a restoration — which is what makes the repeat measurable, and what makes it necessary. The pre-registered five-seed campaign in Appendix C is designed to settle the effect size; what this paper establishes without it is the ordering of the arms and the mechanism that produces it, since the three controls sit at zero while only the arm carrying A_H moves.

The effective horizon is one step. The calibration in Eq. (19) suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is dominated by a single step ahead. This is the predicted behavior of Eq. (18) rather than a bug, but it means the “future” in the visitation term is a short future, and a forecaster whose error did not grow with k would test the idea more sharply. A sequential forecaster is the obvious comparison and we have not run it.

The support is small by construction. Section 9 bounds the fraction of positions carrying a non-zero A_H at $2(n-1)/T$, on the order of a percent for our settings. Whether the effect scales with n , or with a branching schedule chosen adaptively rather than at fixed depths, is untested.

An assumption with no remedy. Approximation B2 — that a token’s update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method rather than introduced here, but our construction adds a second estimator that relies on it.

One prediction of the analysis is unmeasured. Section 4.4 argues that the two channels are anti-correlated. The prefactor half of that claim is analytic and the A_H half is inferred from a correlation with branch diversity; we have not measured their *product* at real sibling-divergence positions in a live rollout. Until that measurement exists, Table 1 is a prediction rather than an observation.

Scope of the backbone evidence. The full decomposition—the five arms, the ablations, the mechanism diagnostics—rests on one 1.5B model. The three further backbones (Section 12.11) carry the three-way contrast at one seed each, which establishes the sign and not its size, and they stop at 7–8B: we have no evidence about models an order of magnitude larger, where the base policy is sharper and the relative size of the two channels may differ. The non-mathematics backbones are also selected on a different benchmark from the mathematics ones (Section 11.2); that is what makes their rows measurable, and it is also why no claim in this paper compares accuracy across backbones.

Ethics Statement

This work studies an optimization-time regularizer for reinforcement learning on mathematics prob-

lems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our method inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the method (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Section 12.12 so that it can be weighed against the benefit.

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whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$. The sign of H' is $-\text{sign}(\delta)$, so any perturbation lowers the entropy, but only at second order. Table 13 contrasts this with a peaked distribution, where the first-order term does not vanish.

distribution	H_0	Ω	ΔH at $\delta=.1$
uniform, $k=10$	2.303	0	-4.7×10^{-4}
uniform, $k=50$	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	-0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	-0.396	-3.8×10^{-2}

Table 13: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

B Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\text{eq}[i, j, t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t , so that $\text{prefixeq}[i, j, t]$ is true exactly when the two rollouts agree on $y_{\leq t}$. Shifting it one position right gives agreement on $y_{< t}$, which is the sibling relation Eq. (15) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

Alignment. $\hat{H}_{\text{togo}}$ is read at position $t+1$ and moved to position t ; at the final position the value is kept. Omitting this shift scores the state before the branch instead of the branch taken.

Memory. The RL path never materializes head logits. For each head the hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding, reduced to an entropy, and released, so peak memory is $O(\text{chunk} \times V)$ rather than $O(\kappa \times B \times T \times V)$.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method’s on random inputs; the $\lambda = 0$ early return in the combination step is what makes this hold exactly rather than to floating-point tolerance.

C Pre-Registered Analysis

The analysis plan below was committed to the artifact repository before any run of the multi-seed campaign was launched, so that the statistics could not be chosen to fit the outcome. It is reproduced here in full; the file is

`docs/preregistration_5seed.md` in the release.

Unit and primary endpoint. The unit of analysis is the seed, not the validation step. Each run is reduced to one number — its mean acc@32 on AIME24 over steps 40–110 — and arms are paired within a seed.

Contrasts, in the order they are reported. Primary: STEER-F – GRPO. Secondary, in this order: STEER-F – STEER, STEER-F – UNIFORM, STEER-F – PERMUTED. Secondary endpoints: maj@32 , uplift, and the six-benchmark mean. Effect sizes are reported with intervals rather than significance stars, and all five arms are reported whatever the outcome.

Stopping and exclusion. Every run goes to step 110; none is stopped early for looking good or bad. A run is excluded only for mechanical failure — out of memory, a crash, a corrupted checkpoint, or a step-0 validation score outside .030–.055, which would indicate a different initial checkpoint. Exclusions are listed with their reason. A failed run is relaunched with the same seed, never a new one, and no seed is added after results are seen.

Checkpoint selection. Two rules are computed and both reported: the argmax over a run’s validation points, and a fixed step 110. The six-benchmark evaluation uses the argmax. Selecting a maximum over twelve noisy checkpoints is biased upward — in the single-seed data by $+.011$ to $+.017$ per arm — which is why the primary endpoint is a window mean rather than a maximum.

What would falsify the central claim. Stated in advance so it cannot be renegotiated afterwards. If STEER-F – UNIFORM straddles zero across seeds, the forecast’s *value* does not matter and the apparatus explains the effect; that is the claim of this paper and it would be refuted. If STEER-F – PERMUTED straddles zero, the pairing between a branch and its score does not matter, which the derivation says it must. If aggregate entropy orders the arms the same way accuracy does, the dissociation argument of Section 12.5 fails, since it depends on the two orderings coming apart. And if the mean token weight in the STEER-F arm departs from ≈ 0.999 , then A_H is not sum-zero over sibling sets as derived and the implementation is wrong rather than the theory.

Amendment, recorded. The pre-registration assigned three seeds to UNIFORM and PERMUTED and five to the other three arms. The campaign as executed runs all five arms at five seeds. The change raises the power of the two control contrasts and was made before any result of the campaign was seen; it is recorded here rather than silently absorbed.

D Within-Run Stability

Earlier drafts of this work paired the arms at each of the eight validation steps in the converged window and reported a paired t over those eight points. That statistic answers a narrower question than the one in Section 12.3: it asks whether a gap is stable along a single trajectory, not whether it reappears when the run is repeated. Its n counts validation points, which are serially correlated within a run, so it is not an estimate of run-to-run uncertainty and cannot be read as one.

We report it here because it is informative about a different thing — whether a gap is an artifact of one lucky validation point — and because the single-seed version of this manuscript quoted it. Where the two disagree, the seed-level statistic in Section 12.7 governs.

The contrast between the two is the point. Table 14 separates every A_H comparison at t_7 between +2.56 and +7.18, which reads as decisive and is not: the same gap sits at the same order as the run-to-run spread we measure in Section 12.7. A statistic can be this confident about a trajectory and say nothing about a re-run, which is exactly why we moved it out of the main text.

contrast	$\Delta\text{acc@32}$	paired t_7
STEER-F – GRPO	+0.0145	+3.08
STEER-F – STEER	+0.0165	+6.68
STEER-F – UNIFORM	+0.0182	+7.18
STEER-F – PERMUTED	+0.0125	+2.56

Table 14: Within-run stability over the eight converged-window validation points of seed 1. The points are consecutive validations of one run and are therefore serially correlated, so $\text{df} = 7$ is generous and these t values are not an estimate of run-to-run variation; see Section 12.7 for that. Majority-vote versions of the same four rows separate at +5.09 to +5.50.